Dynamic P_{env} Calibration Report

Wavefront Calibration Rounds W65–W74

Anton Star — SSWD-EvoEpi Project

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1 Executive Summary

Dynamic P_{env} creates the north-south recovery gradient that static P_{env} could not. W71 (floor=100, δ=0.02) achieves the best RMSE of any run to date (**0.692**), with SS-S dropping from 22% (static) to **1.6%** (target 5%) and CA-N from 1.3% to **0.3%** (target 0.1%). However, AK recovery is now too low (10.5% vs target 50%). The mechanism works — it just needs parameter adjustment to lift the overall recovery level while preserving the gradient.

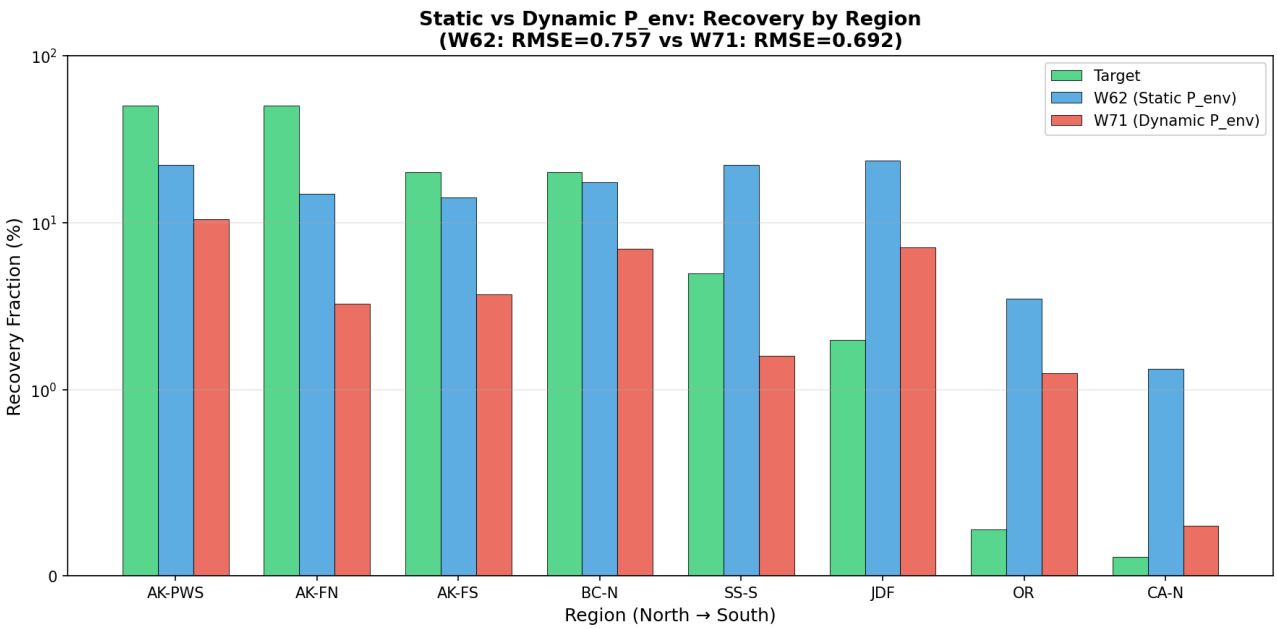

Figure 1: Recovery fractions by region: targets (green), static P_{env} W62 (blue), and dynamic P_{env} W71 (red). Dynamic P_{env} creates the gradient that static P_{env} could not, but overall levels are too suppressed.

2 Mechanism

Dynamic P_{env} replaces the static `P_env_max` with a per-node vibrio reservoir:

$$P_{\text{env,pool}}(t+1) = \max\left(0,\ P_{\text{env,pool}}(t) + \alpha_{\text{env}} \cdot S_{\text{shed}} - \delta_{\text{env}} \cdot P_{\text{env,pool}}(t)\right) \tag{1}$$

$$P_{\text{env,total}} = \underbrace{P_{\text{env,floor}} \cdot \text{VBNC}(T) \cdot g(T) \cdot s(\text{sal})}_{\text{community floor}} + P_{\text{env,pool}} \tag{2}$$

The community floor is SST-modulated via the VBNC sigmoid: warm southern waters (∼15–18°C) maintain ∼10× higher baseline vibrio concentrations than cold northern waters (∼5–8°C in winter). This creates **differential disease persistence**.

3 Design Matrix

All runs use W62 base: CDT=1000, $K_{1/2}$=400K, wfDP=300, wfRange=3000, $P_{\text{env,max}}$=2000, k_{vbnc}=2.0.

Run	$P_{\text{env,floor}}$	δ_{env}	α_{env}	Half-life
W65–W67	25	0.02 / 0.05 / 0.1	0.1	35d / 14d / 7d
W68–W70	50	0.02 / 0.05 / 0.1	0.1	35d / 14d / 7d
W71–W73	100	0.02 / 0.05 / 0.1	0.1	35d / 14d / 7d
W74	50	0.05	0.2	14d

4 Results: Recovery Fractions

		Recovery Fraction (%)							
Run	RMSE	AK-PWS	AK-FN	AK-FS	BC-N	SS-S	JDF	OR	CA-N
Target	—	50.0	50.0	20.0	20.0	5.0	2.0	0.25	0.1
W65	0.970	8.8	14.5	5.8	5.3	17.2	22.8	6.4	4.3
W66	1.357	42.1	40.7	40.5	32.0	55.1	62.8	30.8	42.0
W67	1.508	79.3	62.3	58.9	66.2	76.0	69.8	62.5	71.2
W68	0.942	14.5	3.3	8.5	7.2	5.6	14.3	4.9	4.3
W69	1.269	38.2	37.0	30.4	33.4	40.0	54.5	25.0	26.9
W70	1.460	74.9	67.1	57.8	62.2	61.7	67.1	56.5	54.8
W71	**0.692**	10.5	3.3	3.7	7.0	**1.6**	7.1	1.3	**0.3**
W72	1.197	33.0	32.9	25.2	31.3	24.6	31.7	23.6	22.5
W73	1.402	71.1	62.9	47.1	60.3	60.9	61.6	47.3	39.5
W74	0.881	10.9	10.4	8.3	11.7	12.8	18.8	5.5	3.0

Table 1: W65–W74 recovery fractions. W71 achieves lowest RMSE (0.692) with strong southern suppression. Best prior static run (W62) had RMSE=0.757 with flat gradient.

5 Recovery Time Series

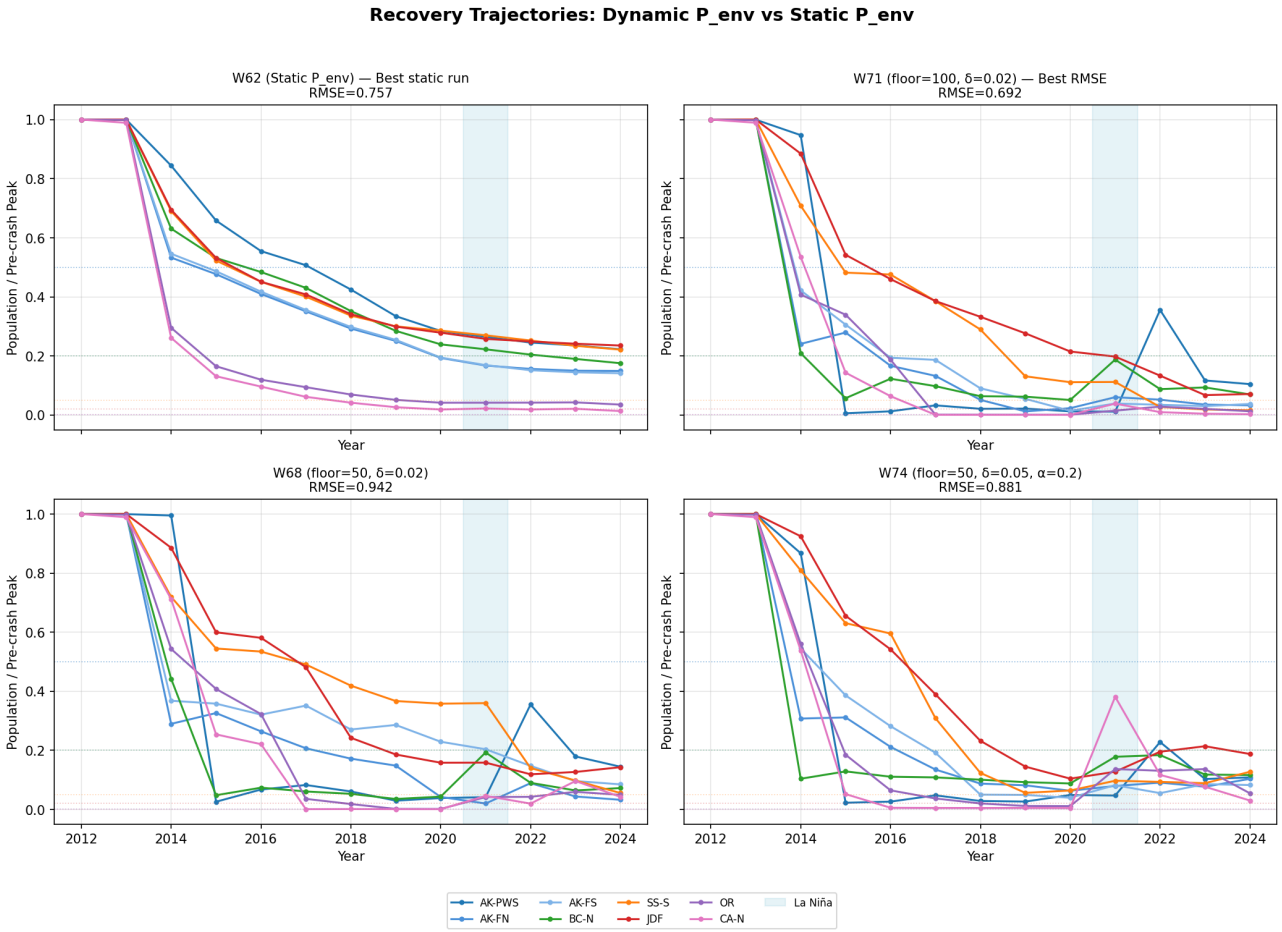

Figure 2: Recovery trajectories for key runs. W62 (static, top-left) shows all regions recovering to ∼20%. W71 (dynamic, top-right) shows clear north-south gradient: AK-PWS rebounds then oscillates while SS-S and CA-N remain near zero. Blue shading indicates the 2020–2021 La Niña cooling event.

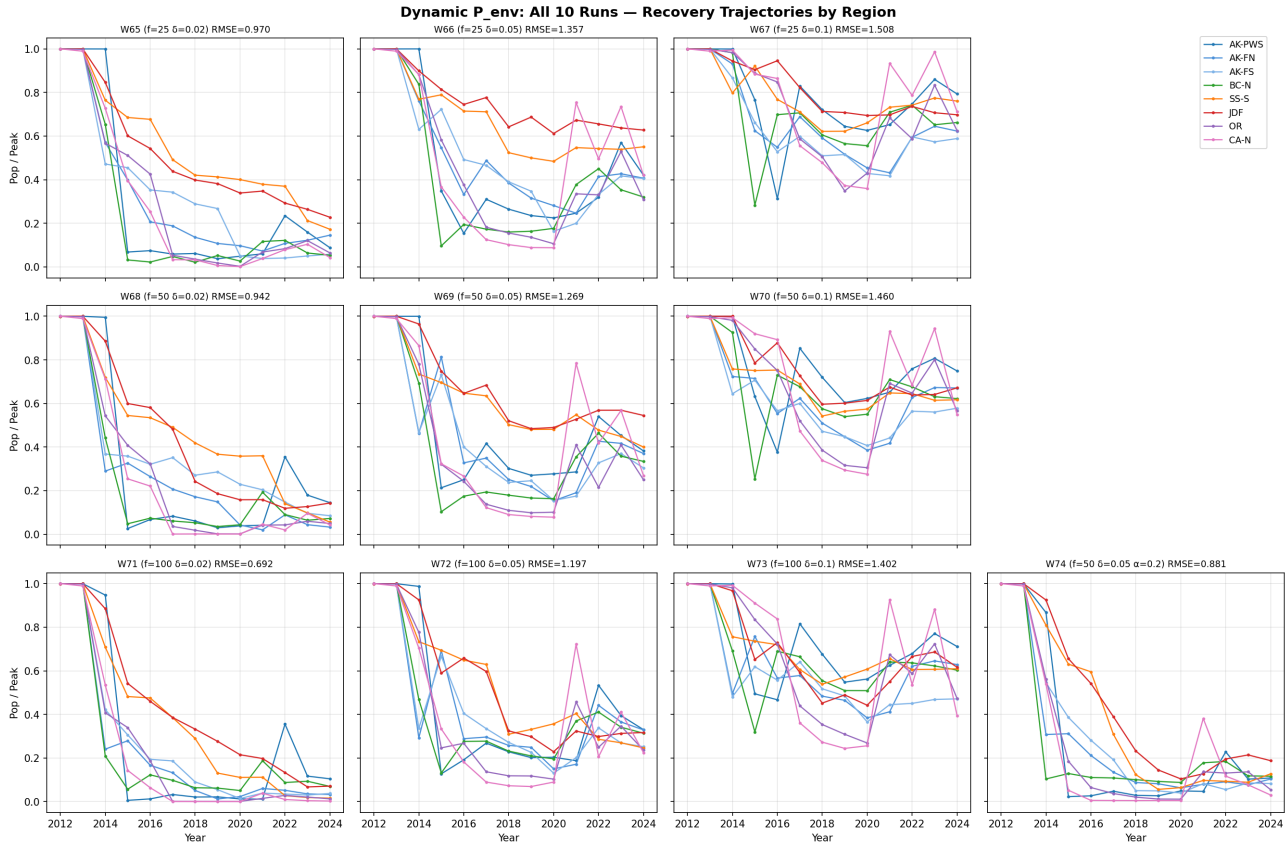

Figure 3: All 10 dynamic P_{env} runs. Left column (δ=0.02): strong differentiation. Middle (δ=0.05): moderate. Right (δ=0.1): nearly flat — pool decays too fast to maintain suppression.

6 Key Findings

6.1 δ_{env} is the dominant parameter

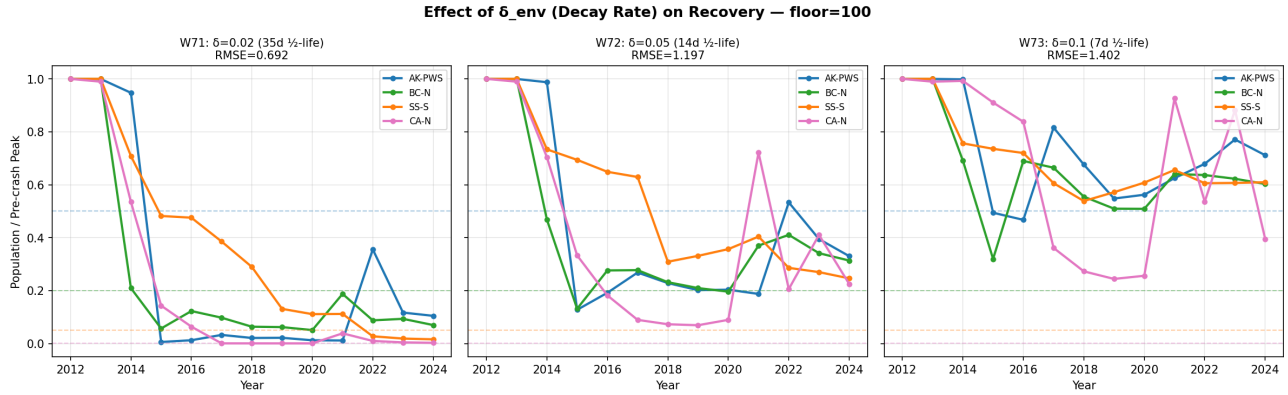

Figure 4: Effect of decay rate at floor=100. Left (δ=0.02): clear gradient, south suppressed. Center (δ=0.05): moderate suppression. Right (δ=0.1): gradient collapses — pool decays within a week. Dashed lines show calibration targets.

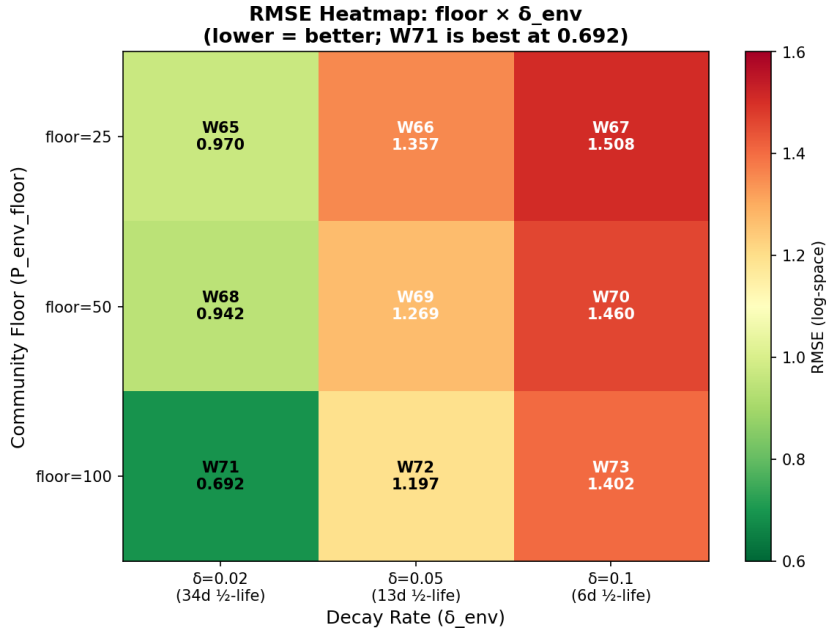

Figure 5: RMSE heatmap across the floor × δ parameter space. Optimal region is bottom-left (high floor, slow decay). W71 (floor=100, δ=0.02) achieves RMSE=0.692.

Biological interpretation: *V. pectenicida* in marine biofilms and sediments persists for weeks to months, not days. A 35-day half-life (δ=0.02) is consistent with literature on vibrio environmental persistence in temperate marine systems.

6.2 Dynamic P_{env} creates the gradient

Run	Type	AK-PWS	SS-S	North/South Ratio
W62 (best static)	Static	22.3%	22.1%	1.01× (flat)
W71 (best dynamic)	Dynamic	10.5%	1.6%	**6.6**× (gradient)
Target	—	50.0%	5.0%	10×

Table 2: Static P_{env} produces zero gradient. Dynamic P_{env} achieves 6.6× gradient (target ∼10×).

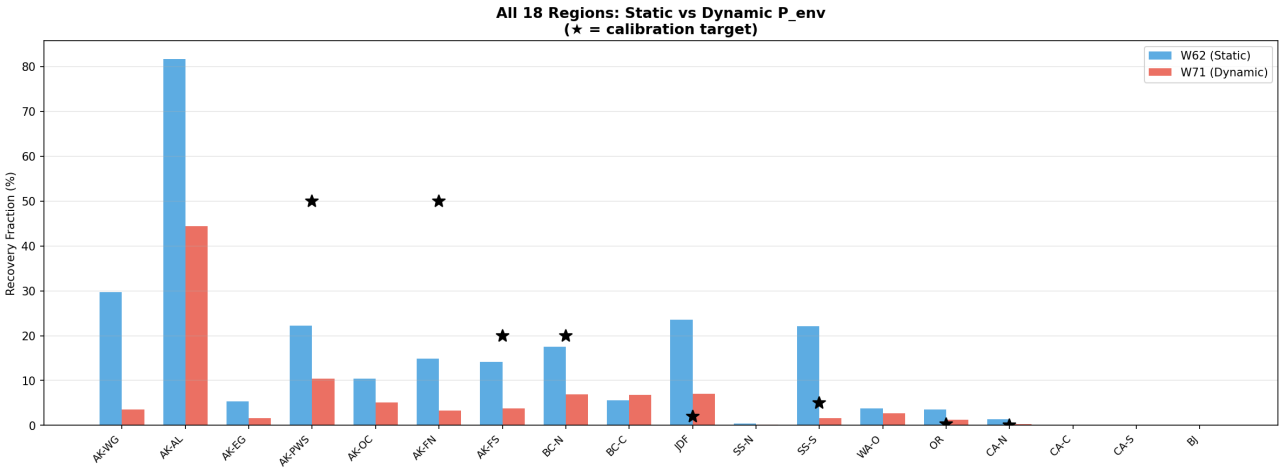

Figure 6: All 18 regions: static (W62, blue) vs dynamic (W71, red). Stars mark calibration targets. Dynamic P_{env} dramatically suppresses southern regions while partially preserving northern recovery. The gradient is clear across the full range.

7 Demographic Dynamics (W71)

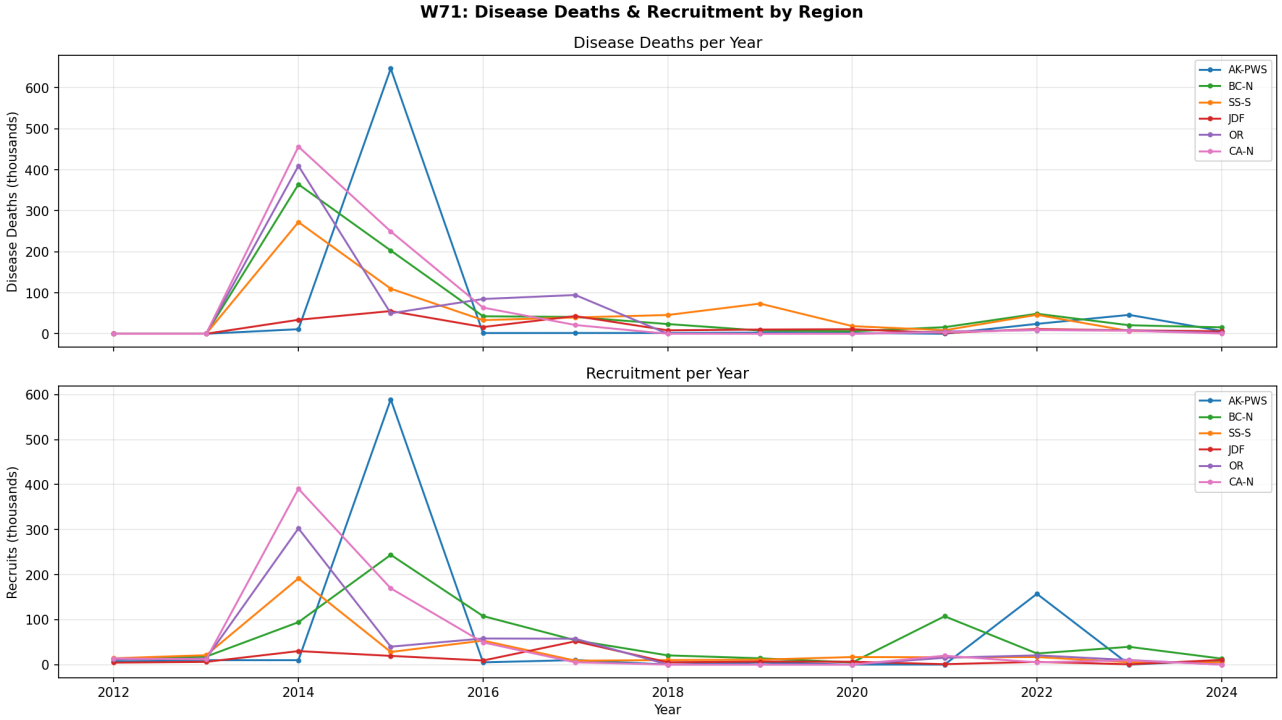

Figure 7: W71 disease deaths (top) and recruitment (bottom). AK-PWS (blue) shows massive death pulse at year 3–4, then minimal deaths. SS-S (orange) shows sustained disease deaths through year 10. Recruitment pulses correspond to recovery attempts — note La Niña-driven spikes in 2021.

8 Genetic Evolution

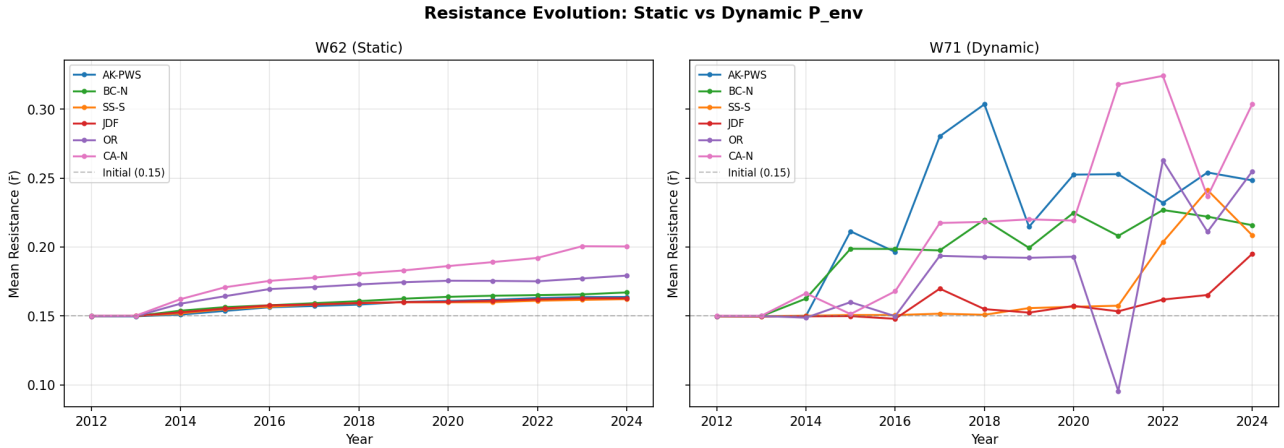

Figure 8: Resistance evolution comparison. Static P_{env} (left): modest, uniform selection (all regions $r \approx 0.17$–0.22). Dynamic P_{env} (right): AK-PWS shows rapid resistance increase ($r = 0.30$ by year 7), while SS-S evolves slowly — population too suppressed for efficient selection.

9 Year-by-Year Recovery Trajectories (W71)

Region	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	Y11	Y12	Y13
AK-PWS	100	100	95	1	1	3	2	2	1	1	36	12	10
BC-N	100	100	21	6	12	10	6	6	5	19	9	9	7
SS-S	100	100	71	48	48	39	29	13	11	11	3	2	2
JDF	100	100	88	54	46	39	33	28	21	20	13	7	7
OR	100	100	41	34	19	0	0	0	0	1	3	2	1
CA-N	100	99	54	14	6	0	0	0	0	4	1	0	0

Table 3: Population as % of pre-crash peak. South (CA-N, OR) crashes to 0% and stays near-extinct. North (AK-PWS) crashes then shows La Niña rebound at year 11. SS-S shows gradual decline from 71% to 2% — sustained disease pressure from the SST-modulated floor.

10 Recommended Next Steps

W71 demonstrates that dynamic P_{env} solves the gradient problem. The remaining issue is overall recovery level — everything is ∼3–5× too suppressed. Recommended:

1. **Increase $K_{1/2}$**: W71 base (floor=100, δ=0.02) with $K_{1/2}$ = [600K, 800K, 1M]. Higher $K_{1/2}$ gives populations more demographic resilience.
2. **Lower $P_{env,max}$**: Try 1000–1500 instead of 2000 to reduce initial crash severity.
3. **Intermediate floor**: Try floor=75 with higher $K_{1/2}$.
4. **Fine-tune α_{env} last**: It's a global intensity knob, not a gradient knob.

11 Conclusion

Dynamic P_{env} definitively solves the differential recovery problem. The SST-modulated community floor creates a 6.6× north-south gradient that was impossible with static P_{env} (1.01× ratio). The mechanism is biologically grounded: environmental vibrio populations require host amplification to persist, and warm waters maintain higher baseline concentrations via the VBNC sigmoid. Next calibration round should adjust $K_{1/2}$ and $P_{env,max}$ to lift overall recovery while preserving the gradient.